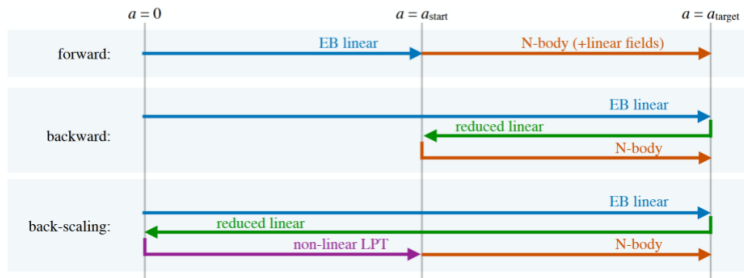


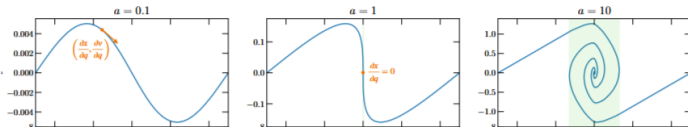
N-body simulations

Angad Anargha Aashish Anjani Sovinity

- The physics governing the infant phase of the Universe, that is dominated by a hot plasma tightly coupled to radiation and linked through gravity to dark matter and neutrinos, is considerably more complex than the purely geodesic evolution of collisionless gravitationally interacting particles [6].
- Since density fluctuations are small, this phase can be treated accurately by perturbative techniques at leading order. State-of-the-art linear-order Einstein–Boltzmann (EB) codes that numerically integrate these coupled multi-physics systems are e.g., CAMB [7] and CLASS [8, 9].
- These codes usually evolve at least dark matter, baryons, photons and (massive) neutrinos and output Fourier-space transfer functions for density δ_X and velocity divergence θ_X for each of the species X as well as the total matter density fluctuations at specifiable output times.



$$\tilde{\delta}_m(k; z_{\text{start}}) = \frac{D_+(z_{\text{start}})}{D_+(z_{\text{target}})} \tilde{\delta}_m^{\text{EB}}(k; z_{\text{target}}).$$



$$\mathbf{x}(\mathbf{q}, \tau) = \mathbf{q} + \Psi(\mathbf{q}, \tau)$$

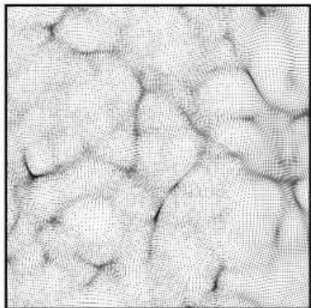
$$\Psi(\mathbf{q}, \tau) = \sum_{j=1}^n D_+(\tau)^j \Psi^{(j)}(\mathbf{q}).$$

$$\Psi^{(2)} = \nabla \phi^{(2)} \quad \text{with} \quad \phi^{(2)} = -\frac{3}{14} \nabla^{-2} \left[\phi_{,ii}^{(1)} \phi_{,jj}^{(1)} - \phi_{,ij}^{(1)} \phi_{,ij}^{(1)} \right],$$

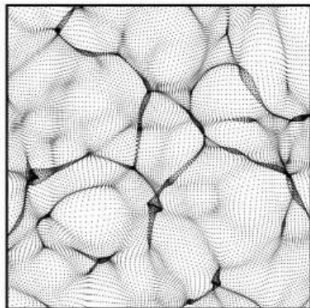
Lagrangian Perturbation Theory

- Higher-order perturbative solutions of Lagrangian displacement were derived. Irrotational second-order solutions (PZA) were derived by Bouchet et al. and Buchert and Ehlers, and third-order solutions (PPZA) were obtained by Buchert, Bouchet et al., and Catelan [1].
- Cosmological N-body simulations show that pancakes, skeletons, and clumps remain during evolution. However, when we continue applying the solutions of ZA, PZA, or PPZA after the appearance of caustics, the nonlinear structure diffuses and breaks.

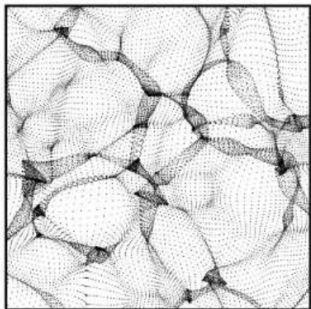
$\sigma=0.5$



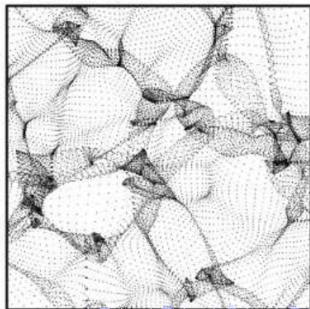
$\sigma=1.$



$\sigma=1.5$



$\sigma=2.$



Lagrangian Perturbation Theory

- Further attempts [1]
 - Adhesion Approximation adds small viscosity term that acts on the high density regions and prevents caustics.
 - Truncated ZA introduces a small scale cutoff to the initial power spectrum as the it is the small scale structure that collapses and forms caustics.
 - Bagla and Padmanabhan developed a new approximation scheme called frozen flow that improved ZA.

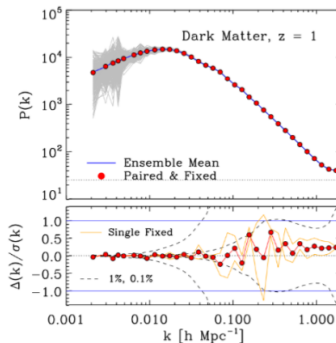
Initial Conditions

$$\phi(\mathbf{x}) = \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} d^3k e^{i\mathbf{k} \cdot \mathbf{x}} \tilde{\varphi}(k) \tilde{W}(\mathbf{k}),$$

$$\langle \tilde{\phi}(\mathbf{k}) \overline{\tilde{\phi}(\mathbf{k}')} \rangle = (2\pi)^3 P(k) \delta_D(\mathbf{k} - \mathbf{k}').$$

$$\tilde{W}(\mathbf{k}) =: \tilde{A}(k) e^{i\theta(k)}$$

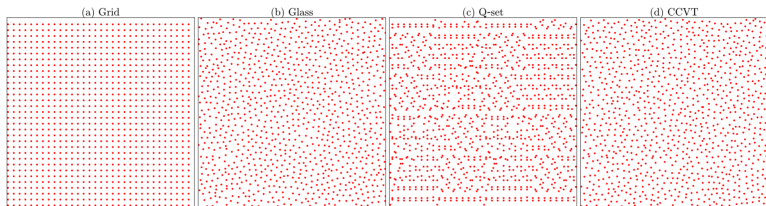
$$\langle \tilde{W}(k) \overline{\tilde{W}(k')} \rangle = \langle \tilde{A}(k) \tilde{A}(k') \rangle = \delta_D(k - k'),$$



Initial Conditions

- The resulting field has much reduced degrees of freedom and a very specific non-Gaussian character. In principle, this introduces a bias in the nonlinear evolution of such a field, e.g., the ensemble averaged power spectrum at $z = 0$ differs from that obtained from an ensemble of Gaussian realizations.
- However, using perturbation theory, [10] showed that the magnitude of this bias in the power spectrum and bispectrum is always smaller (by a factor equal to the number of Fourier modes) compared to mode-coupling terms of physical origin.
- It was found empirically that these simulations produce highly accurate estimates of non-linear clustering (including power spectra, bispectra, halo mass function, and others) and that the level of spurious non-Gaussianity introduced is almost undetectable for large enough simulation boxes [2].

Initial particle loads



N body equations

$$\frac{dX^i}{dt} = \frac{P^i}{m} = \frac{P_i}{ma^2} \quad \text{and} \quad \frac{dP_i}{dt} = -m \frac{\partial \phi}{\partial X^i}$$

$$\frac{\partial f_m}{\partial t} + \frac{p_i}{ma^2} \frac{\partial f_m}{\partial x^i} - m \frac{\partial \phi}{\partial x^i} \frac{\partial f_m}{\partial p_i} = 0$$

$$\nabla^2 \phi = 4\pi G a^2 (\rho - \bar{\rho})$$

$$\rho = \frac{m}{a^3} \int_{\mathbb{R}^3} d^3p \, f_m(\mathbf{x}, \mathbf{p}, t)$$

FORCE CALCULATIONS

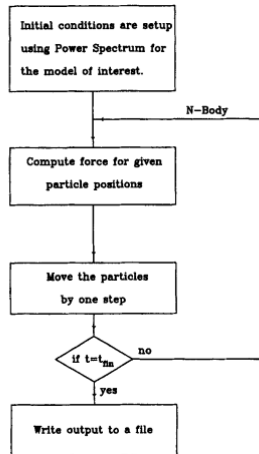


Figure: Flowchart for an N-body code

Setting up the Force Equations

- The evolution of perturbations can be studied in the Newtonian limit, at scales $\ll \frac{c}{H_0}$.
- Comoving volume in phase space

$$\nabla^2 \phi = \frac{3H_0^2 \Omega_m}{2a} \delta \quad (1)$$

$$g(\vec{x}) = -\nabla \phi = \frac{G}{a} \int_{\mathbb{R}^3} \frac{\rho(\vec{x}')(\vec{x} - \vec{x}')}{|\vec{x} - \vec{x}'|^3} d^3 \vec{x}' \quad (2)$$

- First Friedmann equation in matter domination

$$\rho(\vec{x}, t) = \sum_{\vec{n} \in \mathbb{Z}^3} \sum_{i=1}^N M_i W(\vec{x} - \vec{X}_i - \vec{n}L)$$

$\vec{n}$ is the index vector for the 3D replicas of the box.

- Appropriate choice of W and large $N \rightarrow$ collisionless dynamics.

Calculating the Force: Brute Force

$$\phi(\vec{X}_i) = -a^{-1} \sum_{\vec{n} \in \mathbb{Z}^3} \left[\sum_{j=1, i \neq j}^N \frac{GM_j}{|\vec{X}_i - \vec{X}_j - \vec{n}L|} + \phi_{\text{box},L}(\vec{X}_i - \vec{n}L) \right]$$

- $\phi_{\text{box},L}$ is sourced by a background $\bar{\rho}$
- **P2P** (particle-to-particle) method.

But how does a computer read this?

Here's how a *time-step calculation* would look like:

```
for i = 1 to  $N_p$  do:
```

```
     $f_i = 0$ 
```

```
Accumulate forces:
```

```
for i = 1 to  $N_p - 1$  do:
```

```
    for j = i + 1 to  $N_p$  do:
```

```
         $f_i = f_i + f_{j \rightarrow i}$ 
```

```
         $f_j = f_j - f_{i \rightarrow j}$ 
```

```
Integrate equation of motion
```

```
for i = 1 to  $N_p$  do:
```

```
     $v_i = v_i + (F_i/m_i)dT$ 
```

```
     $x_i = x_i + v_i dT$ 
```

```
Update time
```

```
 $t = t + dT$ 
```


P2P

Problems:

- $\mathcal{O}(N^2)$
- Double sum converges slowly
- Truncation errors

Is it a lost cause? No!

- Calculates forces to machine precision.
- Good for short-ranges (Lenard-Jones-like).
- **Standard Ewald summation:**
 - 1 Replace the sum by two independent sums
 - 2 Long range contributions $\rightarrow$ Fourier space
Short range contributions $\rightarrow$ Configuration space

Ewald Summation

- Historically introduced for calculations of Coulombic potentials in ionic crystals
- What is 'short' and what is 'long'?

$$\phi(\vec{x}) = \phi_{lr}(\vec{x}) + \phi_{sr}(\vec{x}) = \mathcal{S} \star \phi + (1 - \mathcal{S}) \star \phi \quad (3)$$

$$S(r; r_s) = \frac{1}{(2\pi r_s^2)^{3/2}} \exp\left(-\frac{r^2}{2r_s^2}\right) \quad (4)$$

- $r < r_s \rightarrow$ short range, $r > r_s \rightarrow$ long range

Using (3), (4) and the linearity of convolutions:

$$\phi = \frac{4\pi G}{a} (\rho - \bar{\rho}) \star \mathcal{G} \quad (5)$$

$$\Rightarrow \mathcal{G}_{lr} \equiv \mathcal{G} \star S \text{ and } \mathcal{G}_{sr} \equiv \mathcal{G} \star (1 - S) \quad (6)$$

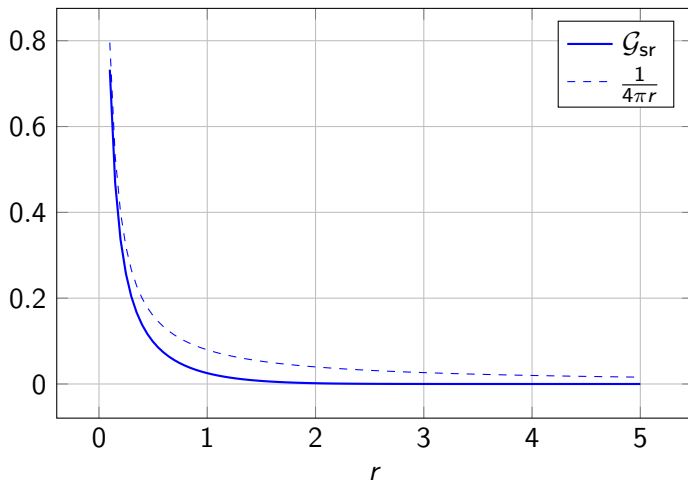
$$\mathcal{G}_{\text{lr}}(r; r_s) = -\frac{1}{4\pi r} \operatorname{erf} \left[\frac{r}{\sqrt{2} r_s} \right], \quad \tilde{\mathcal{G}}_{\text{lr}}(k; r_s) = -\frac{1}{k^2} \exp \left[-\frac{1}{2} k^2 r_s^2 \right] \quad (7)$$

$$\mathcal{G}_{\text{sr}}(r; r_s) = -\frac{1}{4\pi r} \operatorname{erfc} \left[\frac{r}{\sqrt{2} r_s} \right], \quad \tilde{\mathcal{G}}_{\text{sr}}(k; r_s) = -\frac{1}{k^2} \left(1 - \exp \left[-\frac{1}{2} k^2 r_s^2 \right] \right) \quad (8)$$

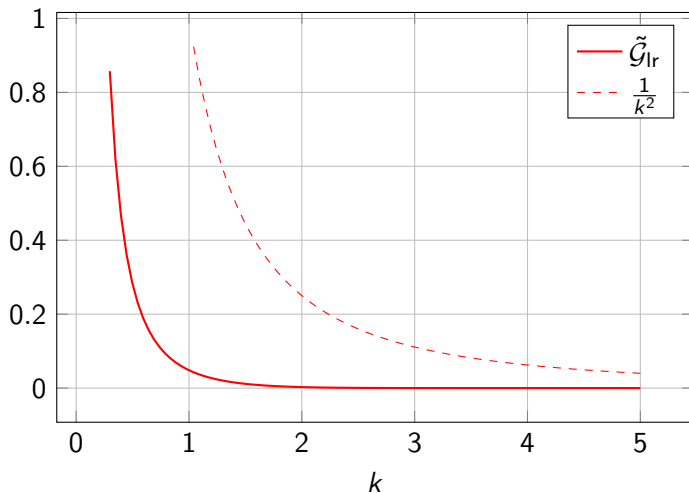
where, $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$ and

$\operatorname{erfc}(x) = 1 - \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt$

Short-range kernels



Long-range kernels



Mesh Methods

- Historically developed by plasma physicists
- Exploits the periodicity of the box
- Sets up a grid throughout the box and calculates ϕ *on the grid*
- PBCs ensure that ρ and ϕ are periodic in $[-L/2, L/2)$ and:

$$\rho(\vec{x}) = \sum_{\vec{n} \in \mathbb{Z}^3} \tilde{\rho} \exp(ik_0 \vec{x} \cdot \vec{n}) \quad (9)$$

$$\phi(\vec{x}) = \sum_{\vec{n} \in \mathbb{Z}^3} \tilde{\phi} \exp(ik_0 \vec{x} \cdot \vec{n}) \quad (10)$$

with $k_0 = \frac{2\pi}{L}$

- Workflow:
 - 1 Assign mass to mesh $\rightarrow$ mass assignment schemes
 - 2 Solve the field equation on the mesh $\rightarrow$ Spectral calculations
 - 3 Calculate the mesh-defined force field $\rightarrow f = -\nabla\phi$
 - 4 Interpolate to find forces on the particles

Mass Assignment Schemes as Cloud Shapes

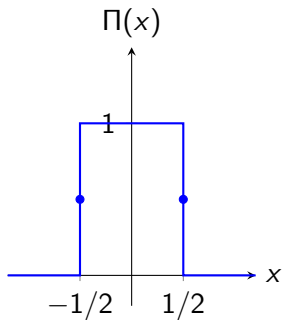
$$\vec{X}_i \rightarrow \vec{x}_{\vec{n}}, \text{ where } \vec{x}_{\vec{n}} = \vec{n}h, \vec{n} \in \mathbb{N}^3$$

Discretisation of mass on the grid:

$$\rho_{\vec{n}} = \int_{\mathbb{R}^3} d^3x' \hat{\rho}(\vec{x}') W_{3D}(\vec{n}h - \vec{x}') \text{ where } \hat{\rho}(\vec{x}) \equiv \sum_{i=1}^N M_i \delta_D(\vec{x} - \vec{X}_i)$$

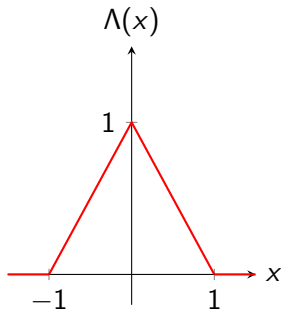
The distribution of overlapping finite-sized particles leads to a smoothly varying continuous mass density, which is sampled at the grid points [11]. Consider a 1-D distribution. Define the *cloud shape* $S(x')$ as the mass density of a particle of unit mass, at a distance x' from its center x_p . Thus,

$$W(x - x_p) = \int_{x_p - h/2}^{x_p + h/2} S(x' - x_p) dx'$$



Define:

$$\Pi(x) = \begin{cases} 0 & |x| > \frac{1}{2} \\ \frac{1}{2} & |x| = \frac{1}{2} \\ 1 & |x| < \frac{1}{2} \end{cases}$$



Define:

$$\Lambda(x) = \begin{cases} 0 & |x| > 1 \\ 1 - |x| & |x| \leq 1 \end{cases}$$

We need to connect S and W .

$$W(x) \equiv \int \Pi\left(\frac{x'}{h}\right) S(x' - x) dx'$$

$$W(x) = \Pi\left(\frac{x}{h}\right) \star S(x)$$

Using the above, we can generate a hierarchy of mass assignment schemes:

$$\text{NGP: } W(x) = \Pi\left(\frac{x}{h}\right) = \frac{1}{h} \Pi\left(\frac{x}{h}\right) \star \delta_D\left(\frac{x}{h}\right) \quad (11)$$

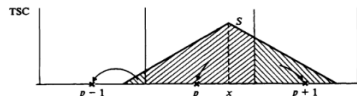
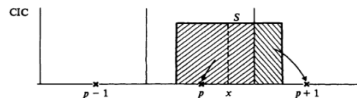
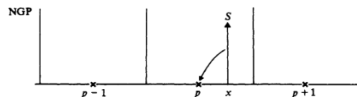
$$\text{CIC: } W(x) = \Pi\left(\frac{x}{h}\right) \star \Pi\left(\frac{x}{h}\right) = \frac{1}{h} \Pi\left(\frac{x}{h}\right) \star \Lambda\left(\frac{x}{h}\right) \quad (12)$$

$$\text{TSC: } W(x) = \Pi\left(\frac{x}{h}\right) \star \Pi\left(\frac{x}{h}\right) \star \Pi\left(\frac{x}{h}\right) = \frac{1}{h} \Pi\left(\frac{x}{h}\right) \star \Lambda\left(\frac{x}{h}\right) \star \Lambda\left(\frac{x}{h}\right) \quad (13)$$

and so on ...

Table 5-1 The hierarchy of assignment schemes of the product form ($H = 1$)

Scheme	Order	Number of points	Cloud shape	Assignment function shape	Force
NGP	0	1^d	δ	Π	Stepwise
CIC	1	2^d	Π	$\wedge = \Pi \cdot \Pi$	Continuous piecewise linear
TSC	2	3^d	$\wedge$	$\Pi \cdot \Pi \cdot \Pi$	Continuous value and first derivative
PQS	3	4^d	$\wedge \cdot \Pi$	$\Pi \cdot \Pi \cdot \Pi \cdot \Pi$	Continuous value first and second derivative

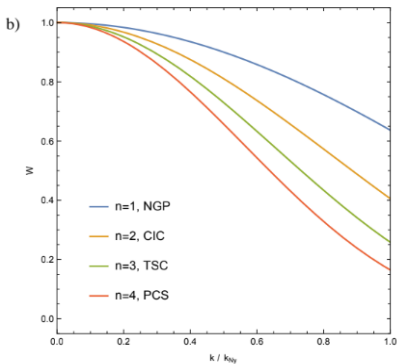
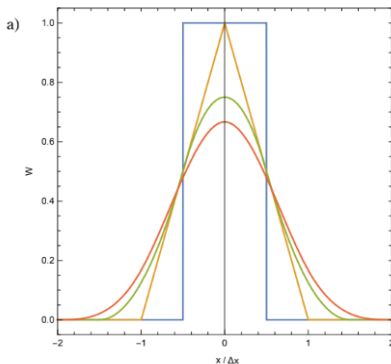
**Figure 5-4** The cloud shape interpretation of charge assignment. The fraction of charge assigned from a particle at position x to a given mesh point is equal to the area of overlap of the cloud shape S with the cell containing that point.

$$W_{\text{NGP}}(x) = \frac{1}{h} \begin{cases} 1 & \text{for } |x| \leq \frac{\Delta x}{2} \\ 0 & \text{otherwise} \end{cases}$$

$$W_{\text{CIC}}(x) = \frac{1}{h} \begin{cases} 1 - \frac{|x|}{\Delta x} & \text{for } |x| < \Delta x \\ 0 & \text{otherwise} \end{cases}$$

$$W_{\text{TSC}}(x) = \frac{1}{\Delta x} \begin{cases} \frac{3}{4} - \left(\frac{x}{\Delta x}\right)^2 & \text{for } |x| \leq \frac{\Delta x}{2} \\ \frac{1}{2} \left(\frac{3}{2} - \frac{|x|}{\Delta x}\right)^2 & \text{for } \frac{\Delta x}{2} \leq |x| < \frac{3\Delta x}{2} \\ 0 & \text{otherwise} \end{cases}$$

- The area of overlap of the cloud shapes with a cell determines the fraction of the charge assigned to the mesh point in that cell.
- Acceleration:
 - NGP $\rightarrow$ piecewise constant (not continuous)
 - CIC $\rightarrow$ piecewise linear (continuous but not differentiable)
 - TSC $\rightarrow$ piecewise quadratic (continuous and first differentiable, but not second differentiable)



Calculating Forces

- Solve the Poisson equation in Fourier space, with periodic boundary conditions.

$$\tilde{\phi}(\vec{k}) = G(\vec{k})\tilde{\delta}(\vec{k})$$

- PBCs require that only $k = \frac{2\pi}{L}$ mode contributes
- But this is automatically captured by the Fast Fourier Transforms (FFTs) — more on this later
- An appropriate Green's function must be used — why does this arise?
- $f = -\nabla\phi$ at grid points and then interpolate to particle positions

On Green's functions

■ Poor man's Green's function: $G(\vec{k}) = -\frac{1}{k^2}$

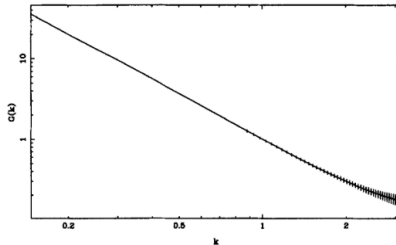
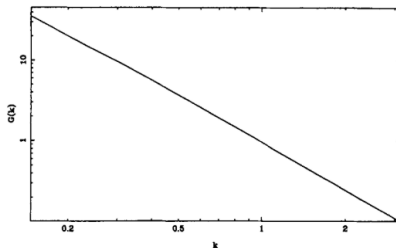
- The physical Green's function
- Isotropic
- Isn't periodic, can lead to **aliasing**, as k modes aren't restricted.

■ Periodic Green's function:

$$G(\vec{k}) = -\frac{L^2}{4(\sin^2(k_x L/2) + \sin^2(k_y L/2) + \sin^2(k_z L/2))}$$

- Green's function of the finite-difference version of the Laplacian
- Periodic
- Anisotropic

Comparing the Green's functions



Taking gradients on a grid

- **FFTs:** $f'_k = -ikf_k$ for any f_k . Errors $\rightarrow$ FFT routine.
- **Real Space Calculations:** Finite differences:

$$\frac{\partial f}{\partial x} = \frac{f(x+h) - f(x-h)}{2h} + \mathcal{O}(h^2)$$

Can be readily extended to higher accuracy by Taylor expansions (alternatively method of undetermined coefficients):

$$\frac{\partial f}{\partial x} = \frac{f(x-2h) - f(x+2h) + 8(f(x+h) - f(x-h))}{12h} + \mathcal{O}(h^5).$$

And even more ...

$$D_x(i, j, k)\phi = \frac{1}{6}(\phi_{i-1,j,k} - \phi_{i+1,j,k}) + \frac{1}{12}(\phi_{i-1,j+1,k} - \phi_{i+1,j+1,k} + \phi_{i-1,j-1,k} - \phi_{i+1,j-1,k} + \phi_{i-1,j,k+1} - \phi_{i+1,j,k+1} + \phi_{i-1,j,k-1} - \phi_{i+1,j,k-1}) \quad (14)$$

The Good, The Bad ...

- And even $\mathcal{O}(N \log N)$
- Periodic boundary conditions are naturally incorporated
- Evades most of the problems of P2P methods

However,

- Resolution limited by grid size
- Forces are often anisotropic

Is that it?

Definitely not!

- Adaptive Mesh Refinement (AMR): Uses PM as a base level, and then refines the grid in high density regions for better resolution.
- Hybrid methods:
 - P^3M ((Particle-Particle)-(Particle-Mesh))
 - TreePM

time evolution: motivation

- Hamiltonian evolution in phase is very restricted. We have quantities that remain invariant throughout the time evolution. These general quantities are sum of objects called poincaré integrals. Intutively this means the sum of projections on any set of hypervolumes which together combine to form the entire phase space is preserved throughout the evolution.
- Any numerical rule of updating positions and velocities of a hamiltonian system must respect these rules to exhibit reliable long term behavior.
- the Runge-Kutta methods are not symplectic. That's why when used to calculate kepler orbits the orbits spread (secular evolution)

time evolution: motivation

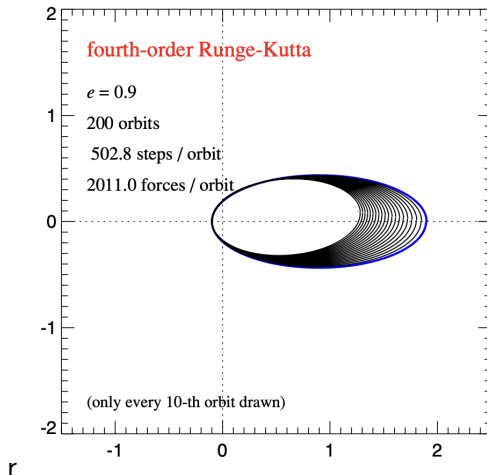


Figure: finding kepler orbits using numerical integration

drifts and kicks

- A general kind of hamiltonian that we deal with is of the form

$$H = H_{kin} + H_{pot}$$
- the following operators are symplectic because they are equivalent to evolving the system under the hamiltonians H_{kin} and H_{pot} respectively.

$$D_t(\Delta t) : \begin{aligned} p_i &\rightarrow p_i \\ x_i &\rightarrow x_i + \frac{p_i}{m_i} \int_t^{t+\Delta t} \frac{dt}{a^2} \end{aligned}$$

$$K_t(\Delta t) : \begin{aligned} x_i &\rightarrow x_i \\ p_i &\rightarrow p_i + f_i \int_t^{t+\Delta t} \frac{dt}{a} \end{aligned}$$

$$f_i = - \sum_j m_i m_j \partial_{x_i} \phi(x_{ij})$$

formalism

- Given a Hamiltonian H , the canonical positions and momentum evolve according to the Hamilton equations:
 $\dot{\xi}_i = \{\xi_i, H\}$ where $\xi = x, p$.
- when the hamiltonian involves involves two parts that depends either on positions or on momenta $H = T(\{p_i\}) + V(\{x_i\})$ then $\dot{\xi} = \{\xi, T\} + \{\xi, V\}$. The first term 'drifts' the system and the second term 'kicks' the system.
- formal solution of a first order operator equation involves the time ordered operator $\xi(t) = \mathcal{T} \exp\left(\int_0^t H(t') dt'\right) \xi(0)$.
- the time ordered integrator for a small interval Δt can be written approximately as a product of drift and kick operators within some order of error in Δt .

drifts and kicks

- we can combine the operators $D_t(\Delta t)$ and $K_t(\Delta t)$ to produce integrators listed below that integrates a hamiltonian very close to the original hamiltonain: $H = H_{kin} + H_{pot} + H_{error}$ where the error is of second order i.e. $\mathcal{O}(\Delta t^2)$

$$D(\Delta t/2)K(\Delta t)D(\Delta t/2)$$

$$K(\Delta t/2)D(\Delta t)K(\Delta t/2)$$

- If the error Hamiltonian is a sufficiently small perturbation, then there exists theorems in dynamical systems ³ that guarantee that almost all orbits that are stable in the real system will continue to be stable in the numerical system. The error does not built up during the course of evolution but it just oscillates.

³for instance the KAM theorem

drifts and kicks

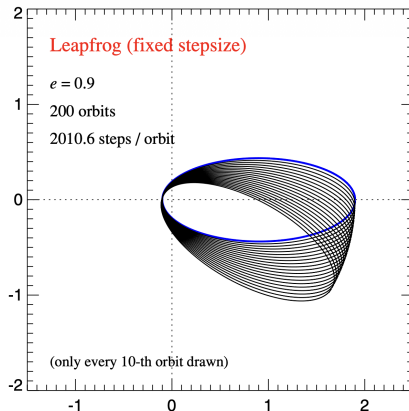


Figure: leapfrog DKD method. the rotation of the ellipse as the time progresses is due to the error term [?].

adaptive time steps

- In cosmological simulations we are confronted with a large dynamic range of time scales. In regions of high mass density, a much smaller time step is required than regions of low density.
- Evolving all the particles with the smallest required time step implies a substantial waste of computational resources.
- The principal idea is to compute forces only for a certain group of particles in a given kick operation (since force calculation is the most expensive part) with other particles being evolved on larger time steps and being kicked more rarely.

hierarchical time integration

- we subdivide the simulation timespan into a power of two hierarchy, then require that the permissible timesteps are nested within this hierarchy. This means that all timesteps are power of two multiples of the smallest occurring timestep.
- Particles with same timestep size occupy a certain timestep bin in this scheme and may change to shorter or longer timesteps when they have completed the time step.
- Recall the second order operator split to integrate a hamiltonian system of the form $H = H_1 + H_2$ is

$$E(H, \Delta t) \approx E(H_1, \Delta t/2) \circ E(H_2, \Delta t) \circ E(H_1, \Delta t/2)$$

hierarchical time integration

- Consider a particle system P described by $H = H_{kin} + H_{pot}$. We can subdivide the particle system into a "slow" and a "fast" part $P = S + F$ based on the notion that slow system S contains particles for which an integration with the given timestep would be sufficiently accurate while the fast system F gets the particles where the integration time step should be chosen smaller.

$$H = H^S + H^F + H_{pot}^{SF}$$

$$E(H, \Delta t) = E(H_{pot}^{SF}, \Delta t/2) \circ$$

$$E(H^F, \Delta t/2) \circ E(H^S, \Delta t/2) \circ E(H^F, \Delta t/2) \\ \circ E(H_{pot}^{SF}, \Delta t/2)$$

- each term in H_{pot}^{SF} will involve one of the slow particles, so $\Delta t/2$ timestep is sufficient for its integration.

hierarchical time step

- But the time step of H^F integrator needs to be reduced.
- define kick K_B^A as the forces due to particles in set B are applied to particles in A .

$$\begin{aligned}
 & K_S^F(\Delta t/2) K_F^S(\Delta t/2) \\
 & E(H_F, \Delta t/2) \\
 E(H, \Delta t) = & K_S^S(\Delta t/2) D_S(\Delta t) K_S^S(\Delta t/2) \\
 & E(H_F, \Delta t/2) \\
 & K_F^S(\Delta t/2) D_S(\Delta t) K_S^F(\Delta t/2)
 \end{aligned}$$

- Notice here that K_S^S commutes with $E(H_F, \Delta/2)$ because they act on different set of particles. Also individual kick operators commute with each other.

hierarchical time step

- then by definition of K_P^P we get

$$K_S^F(\Delta t)K_F^S(\Delta t)K_S^S(\Delta t) = K_P^P(\Delta t)K_F^F(-\Delta t).$$

So the last expression becomes

$$\begin{aligned} & K_P^P(\Delta t/2)K_F^F(-\Delta t/2) \\ E(H, \Delta t) &= E(H_F, \Delta t/2)D_S(\Delta t)E(H_F, \Delta t/2) \\ & K_F^F(-\Delta t/2)K_P^P(\Delta t/2) \end{aligned}$$

This is the form of the integrator which is recursively applied in practice.

- First start out by calculating forces for the whole system P. This allows us to apply K_P^P . Now the size of the forces calculated for P can be used to distinguish between F and S particles by means of a time step criterion.

hierarchical time step

- For particles ending up in F , new forces are then calculated sourced by particles in just the set F . This calculation allows us to partially undo the previous K_P^P kick operator with K_F^F .
- The F set is treated as then a system of new particles P' for which we need to calculate integrator over half the original time step: $E(H_F, \Delta t/2)$. This can be dealt with **recursively** by applying the same procedure yet again with P' becoming new P . Notice that the first task of calculating all forces within P' is already just been done in previous step as force calculation for the set F .
- Also the forces entering the time step criterion for deciding between fast and slow particles in the deeper levels of hierarchy are partial forces originating from even smaller subset P' of the system. The recursive subdivision into ever smaller particle sets end once the set F becomes empty.

representation

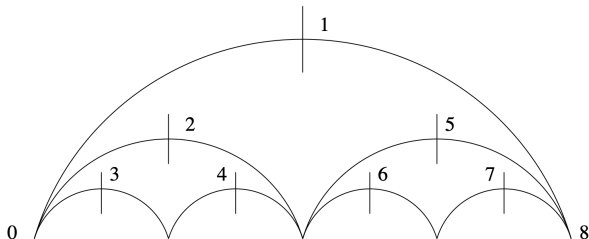


Figure: graphical representation of 3 level hierarchical time stepping. the arcs represent kick operations and vertical lines represent select operation and subsequent drift.

Beyond the Cold Collisionless Limit

- Till now, we have considered all masses in the universe to be cold and collisionless fluid.
- Different candidates of the DM like, WIMPs and SIDM.
- Numerical Consideration and the Challenge of High-Accuracy Simulations

Weakly Interacting Massive Particles

- It has a very cold distribution function because of its early decoupling and high mass.
- Motivated by the observation that if the cross-section was set by the electroweak scale of around 100GeV mass, then it would result in a relic abundance matching the measured density of DM in the universe, known as the WIMP miracle.
- Non-detection of supersymmetric particles (WIMP candidate) at the LHC to-date, and the strong constraints from direct detection experiments begin to challenge the explanation of dark matter through thermally produced WIMP.
- Neutralino (Supersymmetric particle) is a good candidate for WIMP as these are stable, interact weakly, and have mass $> 100\text{GeV}$.

Weakly Interacting Massive Particles

- On an astrophysical scale, neutralinos are like a perfectly collisionless fluid because of small cross section. However, finite temperature decoupling implies that they have a small but non-zero random microscopic velocity.
- As a result, they stream freely out of perturbation of sizes $0.7pc$, which means a low mass halo $< 10^{-8}M_{sun}$ is suppressed.
- For the neutralino field, the cold limit is also a good approximation for the distribution function because thermal random velocities are small compared to the mean velocities from the gravitational instability, specially at late times where the former are cooled due to expansion of the universe whereas the latter keep increasing due to structure formation.

Weakly Interacting Massive Particles

- Consequently, numerical simulations of the structure formation with neutralinos assume them to be perfectly cold, follow them with N-body integration methods, and incorporate their free streaming effects only in the initial power spectrum suppressing the amplitude of the small scale modes. By removing the tiny fluctuations, a cold and warm fluid behaves exactly the same way on large scales.
- The central part of the halos could be affected by the intrinsic neutralino velocity dispersion, providing maximum bounds on the density.
- To resolve the full hierarchy of possible structures, it would require about 10^{23} N body particles. For this reason, neutralino simulations have focused on the formation of the first structures at high redshifts.

Self Interacting Dark Matter

- Any physically motivated particle DM model must have a non-zero interaction cross-section in order for those particles to be produced in the early Universe in the first place.
- There are various possible interaction mechanisms but carried out by exchange of dark photon or scalar. For instance, dark matter can interact and annihilate with itself (e.g., if it were a Majorana particle) producing standard model particles and (gamma-ray) photons.
- This 'indirect detection' of particle dark matter heavily relies on N-body simulations which can pinpoint the most likely places for a putative detection, as well as the expected emission.
- Other possible self-interaction considers the weakly collisional regime with elastic binary processes $\chi\chi \rightarrow \chi\chi$. This case is generically known as self-interacting dark matter, and the crosssections can be large enough (light mediator) to be relevant for structure formation.

Self Interacting Dark Matter

- For instance, weak collisionality was found to have an effect on the density profiles of isolated halos, predominantly by isotropising the core velocity dispersion and reducing the density. In principle, such systems can undergo a gravothermal catastrophe leading to core collapse, but current constraints on the cross-section put the time-scale for this at $> 100 \text{ Gyr}$.
- Self-interacting DM requires a modelling of microscopic short-range particle scattering in N-body simulations. These interactions alter the overall the dynamics which is no longer purely geodesic, and thus it requires upgrading from the Vlasov–Poisson to the Boltzmann Poisson system of equations by adding a scattering balance term.

$$\frac{\partial f}{\partial t} + \frac{\mathbf{p}}{ma^2} \cdot \nabla_x f - m \nabla_x \phi \cdot \nabla_p f = \Gamma_{\text{in}} - \Gamma_{\text{out}}$$

- Where Γ_{in} is the instantenous rate of change of $f(\mathbf{x}, \mathbf{p}, t)$ due to all scattering events that leads to particles ending up in an infestisimal phase space volume $d^3x d^3p$ around $f(\mathbf{x}, \mathbf{p})$ and Γ_{out} the respective rate for scattering out of that volume.

Self Interacting Dark Matter

- Assuming a two-particle momentum 'in state' (p, p_1) is scattered into the 'out state' (p', p'_1) , then the scattering rate balance, given a differential scattering cross section $\frac{d\sigma_\chi}{d\Omega}$, is

■

$$\Gamma_{\text{in}} - \Gamma_{\text{out}} = \int_{\mathbb{R}^3} d^3p_1 \int_0^{4\pi} d\Omega \frac{d\sigma_\chi}{d\Omega} \frac{\|\mathbf{p} - \mathbf{p}_1\|}{ma^2} \\ \times [f(\mathbf{x}, \mathbf{p}'; t)f(\mathbf{x}, \mathbf{p}'_1; t) - f(\mathbf{x}, \mathbf{p}; t)f(\mathbf{x}, \mathbf{p}_1; t)]$$

- Unfortunately, if one inserts the N-body distribution function in the scattering rate $\Gamma := \Gamma_{\text{in}} - \Gamma_{\text{out}}$, one does not arrive at a practicable discretisation since Γ is zero unless two N-body particles are located at the exact same location. Since each N-body particle anyway represents an element of phase space in a coarse-grained sense, practical approaches have to resort to estimating Γ with smeared out particles.

Self Interacting Dark Matter



$$f_N(\mathbf{x}, \mathbf{p}, t) = \sum_{\mathbf{n} \in \mathbb{Z}^3} \sum_{i=1}^N \frac{M_i}{m} W(\|\mathbf{x} - \mathbf{X}_i(t) - \mathbf{n}L\|; h_i) \delta_D(\mathbf{p} - \mathbf{P}_i(t))$$

- Where M_i is mass of smeared out N body particle, m is the particle mass, $\vec{P}_i$ is momentum of the smeared out N body particle and $W(r;h)$ is a smooth kernel function with finite support and h_i a bandwidth or smoothing scale parameter chosen so that several other particles are found within a distance h_i from particle i .
- This distribution function can then be used to express the scattering rate experienced by particle i as

$$\Gamma_i = \sum_j \Gamma_{i|j} = \sum_j \frac{\sigma_\chi}{ma^2} \|\mathbf{P}_i - \mathbf{P}_j\| V_{ij},$$

$$\text{with } V_{ij} = \int_{\mathbb{R}^3} d^3x W(\|\mathbf{x} - \mathbf{X}_i\|; h_i) W(\|\mathbf{x} - \mathbf{X}_j\|; h_j)$$

Self Interacting Dark Matter

- representing the overlap between the particle kernels. Since numerically $\Gamma_{i|j} = \Gamma_{j|i}$ is not guaranteed, Rocha et al. (2013) proposed to use a symmetrised scattering rate $\Gamma_{ij} := (\Gamma_{i|j} + \Gamma_{j|i})/2$. Once the scattering rate Γ_{ij} is known, the scattering probability over a time step Δt is $P_{ij} = \Gamma_{ij}\Delta t$. Then, one can realise an elastic scattering event by sampling this probability over one timestep and randomising the direction of the relative velocity vector of particles i and j in the centre-of-mass frame.
- Another possibility is decaying dark matter, where the dark matter decays into massless species on cosmological time scales. Current constraints from the CMB require that if all the DM decays into photons, then the decay rate is at most $\Gamma^{-1} > 160 \text{ Gyr}$. However, a much smaller fraction of DM can decay and still leave a cosmological signature.
- In this case, the corresponding increase in energy density of relativistic species at late times needs to be accounted for in the background evolution of numerical simulations.

Self Interacting Dark Matter

- There is a priori no reason that the physics of the dark sector could not be significantly richer, with multiple dark species and internal degrees of freedom.
- An intermediate space between self-interacting dark matter and decaying dark matter is occupied by dissipative dark matter, which allows for up-scattering to an excited state $\chi\chi \rightarrow \chi'\chi'$ with a subsequent decay, e.g. $\chi' \rightarrow \chi + X$ under emission of a light or massless particle 'X'.
- Such processes can efficiently remove energy from the centres of halos and lead to halo core collapse in less than a Hubble time, leaving testable signatures in dwarf galaxies.

Box Size and Mass Resolution

- Box Size : Because simulations use a finite box with periodic boundary conditions, any modes larger than the box size (Fourier mode $k < \frac{2\pi}{L}$)

$$\lambda > L \implies \frac{1}{\lambda} < \frac{1}{L} \implies \frac{2\pi}{\lambda} < \frac{2\pi}{L} \implies k < \frac{2\pi}{L}$$

are completely missed.

- This effectively means the simulated region lacks large scale tidal forces and evolves as if it were at the exact cosmic mean density as long wavelength are suppressed. Missing these large modes suppresses the abundances of massive dark matter halos and lowers the amplitude of the power spectrum.

Box Size and Mass Resolution

- Mass Resolution : Finite particle mass determines the smallest dark matter halo that can be successfully resolved (the halo formed from the grouping of many particles, but if the particle mass is huge then we can't resolve smallest DM halo).
- If the simulation fails to resolve small halos that significantly contributed to the non-linear structure, the resulting density spectrum will be artificially biased low.

Close Encounters and Regularization in N body Methods

- In true collisionless DM fluid, DM particles do not physically collide. However in a N body simulation, a single particle actually represents a massive chunk of matter.
- To prevent these artificially massive particles from undergoing unrealistic two body scattering or experiencing infinite acceleration during close encounters, simulations introduce a softening length ϵ .
- The softening length modifies the standard Newtonian gravitational force at short distances so that the force peaks and smoothly falls to zero as the distance between particles approaches 0.

$$\phi_{\text{soft}}(r) = -\frac{GM}{\sqrt{r^2 + \epsilon^2}}$$

$$a_{\text{soft}} = -\frac{GM}{(r^2 + \epsilon^2)^{3/2}}r$$

Close Encounters and Regularization in N body Methods



$$\text{as } r \rightarrow 0, \quad a_{\text{soft}} \rightarrow 0$$
$$\phi_{\text{soft}} \rightarrow -\frac{GM}{\epsilon}$$

The maximum acceleration is reached at $r = \epsilon/\sqrt{2}$ with a maximum value of the acceleration is $2/(3\sqrt{3}\epsilon^2)$.

- Choosing the right softening length is a balancing act: If it is too large, the simulation smooths out real (If too large then if they are so close then they do not even feel strong force so the clump are not tightly held).

The N body Approximation

- Despite the excellent agreement betⁿ diff N-body codes, there is still the fundamental question of whether the N-body discretization itself is a valid approximation of the true collisionless fluid.
- Even after convergence after increasing the resolution by increasing particles and smaller timesteps, it may be not the true collisionless fluid (Vlasov-Poisson equation) solution. This is because most simulation codes rely on the same underlying approximation: representing a continuous dark matter fluid using discrete particles.
- As a result, different simulations may converge to the same solution within this framework, yet still deviate from the true continuum behavior due to discreteness effects such as noise (artificial density fluctuation) and artificial interactions (unphysical forces and energy exchange when particles come too close) wrt collisionless assumption.

Correlation Function and Power Spectrum

- Spatial properties of a statistically homogenous set of points characterized by the N-point coorelation function.
- The two-point correlation function is defined as:

$$\xi(|\mathbf{x} - \mathbf{x}'|, t) \equiv \langle \delta(\mathbf{x}, t) \delta(\mathbf{x}', t) \rangle \quad (15)$$

- $\xi(r) > 0$: matter clustered at that scale
- $\xi(r) = 0$: random distribution
- The Fourier transform of the two-point correlation function is the Power Spectrum, $P(k)$.

$$\langle \delta(\mathbf{k}) \delta^*(\mathbf{k}') \rangle = (2\pi)^3 P(k) \delta_D(\mathbf{k} - \mathbf{k}') \quad (16)$$

- Power spectrum quantifies the amplitude of fluctuations.

Correlation Function and Power Spectrum

- Linear Perturbation Theory states that Fourier modes of small fluctuations evolve independently.
- For non-Gaussian field, all the connected n-point functions are needed to fully describe the field.
- Higher-order correlations are more efficiently estimated and characterized by their volume averages, irreducible moments (cumulants) of counts in cells, $\bar{\xi}_N$.
- Compare statistics of simulated universe to real universe through $P(k)$.

Fast Fourier Transform

- For computational efficiency, the correlations in Fourier space can be estimated employing Fast Fourier Transform (FFT) applied over the density field estimated on a uniform grid.
- FFT scales with the no. of grid points N_g as $O(N_g \log N_g)$
- One FFT algorithm is called the *decimation – in – time* algorithm.
- The *Discrete Fourier Transform* of a sequence x_n is:

$$X_k \equiv \sum_{n=0}^{N-1} x_n e^{-j \frac{2\pi}{N} nk}; \quad k = 0, \dots, N-1 \quad (17)$$

[3]

Fast Fourier Transform

- *FFT butterfly* splits the sum into a sum over even n and sum over odd n :

$$X_k = \sum_{n=0}^{\frac{N}{2}-1} x_{2n} e^{-j\frac{2\pi}{N}(2n)k} + \sum_{n=0}^{\frac{N}{2}-1} x_{2n+1} e^{-j\frac{2\pi}{N}(2n+1)k} \quad (18)$$

for $k = 0, 1, \dots, N - 1$

- Since the one-point DFT requires $\log_2(N)$ stages, the amount of computation required to compute the N -point DFT is $N/2 \log_2(N)$ complex multiplications and $N \log_2(N)$ complex additions.

Halo Finders

- Halo finders take raw data from simulation (positions and velocities of dark matter particles) and organize them into halo catalogues [4].
- Halo catalogues contain dark matter halos or groups of dark matter particles and their intrinsic properties like position, velocity, mass, radius, etc.
- The three most common halo finders currently in use are the classical configuration space **friends-of-friends' (FoF)**, **spherical overdensity (SO)** and **phase-space FoF**.

FoF

- Friends of Friends is an algorithm that groups N-body particles whose maximum separation is below a threshold, referred to as "*linking length*" (l_1) and given in terms of the mean inter-particle separation.
- FoF effectively identifies regions in terms of an isodensity contour with a value:

$$\delta \sim 0.65 \ell_l^{-3} - 1 \quad (19)$$

- For typical values of $\ell_l = 0.2$, this corresponds to ~ 80 times the mean density.
- Spherical Collapse Model: A region of the universe that stops expanding and collapses reaches a stable state when its density is 178 times the mean density of the universe.

SO

- Spherical Overdensity identifies spherical regions whose mean enclosed density is above a threshold, usually matching the value expected from spherical collapse ($200 \bar{\rho}$).
- Finds local peaks (maxima) in the density field.
- **Grid method:** Simulation box is divided into a 3D mesh, and the number of particles in each grid cell is counted. The cell with the highest count is taken as the halo centre.
- **Smoothed Particle Hydrodynamics (SPH) kernel:** For every particle, algorithm calculates a local density by looking at its nearest N neighbours. Particle with the highest local density as the maximum.

Phase-space FoF

- Phase-space FoF (eg. ROCKSTAR) improves FoF to include sub-structures.
- Works in 6D phase space (x, y, z, v_x, v_y, v_z) .
- The basic idea is to split the standard FoF group into a hierarchy of structures recursively identified into a FoF algorithm but with a linking length and distance metric defined in phase space, thus grouping together particles nearby both in position and velocity.
- Distance metric:

$$s^2 = \frac{(\Delta r)^2}{\sigma_r^2} + \frac{(\Delta v)^2}{\sigma_v^2} \quad (20)$$

where σ_r and σ_v are scaling factors

Phase-space FoF

- Distinguish between halos and subhalos: Since the subhalo particles are moving together as a coherent group different from the parent halos.
- Discards particles that are not gravitationally bound: If a particle is flying through a halo at high speed, standard FoF will include its mass.

But the phase-space FoF calculates the potential energy and kinetic energy. If the particles's velocity is higher than the escape velocity, the algorithm discards the particle.

References I



T. Tatekawa, *Lagrangian perturbation theory in Newtonian cosmology*, arXiv:astro-ph/0412025 (2005).

<https://arxiv.org/abs/astro-ph/0412025>



R. E. Angulo and O. Hahn, *Large-scale dark matter simulations*, Living Reviews in Computational Astrophysics **8**(1) (2022).

<https://doi.org/10.1007/s41115-021-00013-z>



S. A. Tretter, *The FFT and Power Spectrum Estimation, in Communication System Design Using DSP Algorithms* (Springer, 2003), pp. 109–124.

https://doi.org/10.1007/978-1-4613-0229-2_4

References II



A. Knebe et al., *Haloed gone MAD: The Halo-Finder Comparison Project*, Monthly Notices of the Royal Astronomical Society **415**(3), 2293–2318 (2011).
<https://doi.org/10.1111/j.1365-2966.2011.18858.x>



M. Davis, G. Efstathiou, C. S. Frenk, and S. D. M. White, *The evolution of large-scale structure in a universe dominated by cold dark matter*, The Astrophysical Journal **292**, 371–394 (1985). <https://doi.org/10.1086/163168>



P. J. E. Peebles, *The large-scale structure of the universe* (Princeton University Press, 1980).

References III



A. Lewis, A. Challinor, and A. Lasenby, *Efficient Computation of Cosmic Microwave Background Anisotropies in Closed Friedmann–Robertson–Walker Models*, The Astrophysical Journal **538**, 473–476 (2000).

<https://doi.org/10.1086/309179>



J. Lesgourgues, *The Cosmic Linear Anisotropy Solving System (CLASS) I: Overview*, arXiv:1104.2932 (2011).

<https://arxiv.org/abs/1104.2932>



D. Blas, J. Lesgourgues, and T. Tram, *The Cosmic Linear Anisotropy Solving System (CLASS) II: Approximation Schemes*, arXiv:1104.2933 (2011).

<https://arxiv.org/abs/1104.2933>

References IV

-  R. E. Angulo and A. Pontzen, *Cosmological N-body simulations with suppressed variance*, Monthly Notices of the Royal Astronomical Society: Letters **462**, L1–L5 (2016).
<https://doi.org/10.1093/mnrasl/slw098>
-  R. W. Hockney and J. W. Eastwood, *Computer Simulation Using Particles* (CRC Press, 1988).
-  C. Lacey and S. Cole, *Merger rates in hierarchical models of galaxy formation – II. Comparison with N-body simulations*, Monthly Notices of the Royal Astronomical Society **271**(3), 676–692 (1994).
<https://doi.org/10.1093/mnras/271.3.676>

References V



B. Diemer and A. V. Kravtsov, *Dependence of the outer density profiles of halos on their mass accretion rate*, The Astrophysical Journal **789**(1), 1 (2014).

<https://doi.org/10.1088/0004-637X/789/1/1>



S. More, A. V. Kravtsov, N. Dalal, and S. Gottlöber, *The overdensity and masses of the friends-of-friends halos and universality of halo mass function*, The Astrophysical Journal Supplement Series **195**(1), 4 (2011).

<https://doi.org/10.1088/0067-0049/195/1/4>